

Library Database — Requirements

Items

- Types: Print Books, EBooks, Magazines, Scientific Journals, Records
- Shared attributes:
 - Item ID
 - Title
 - Creator/Author
 - Publisher
 - Year
 - Subject/Genre
 - Language
- Type attributes:
 - Print Books: ISBN, Page count
 - EBooks: ISBN, Format (PDF, EPUB, online only, etc.)
 - Magazines: ISSN, Issue, Issue date
 - Scientific Journals: ISSN, Issue, Volume
 - Records: Type (Any other item type), Quantity - Are records of items to add
- Copies: Copy is its own entity, weak — identified by its Item plus a copy number
 - Copy number
 - Status (available, on loan, on hold, lost, in repair, reference-only)
 - Every item type has copies, including EBooks, Magazines, Journals and Records
- Relationships:
 - Item — Unit Of — Copy (one Item, many Copies)

Members

- Info tracked per member:
 - Member ID
 - Name
 - Address
 - Phone
 - Email
 - Registration date

- Status (active, expired, suspended)
- Rules:
 - Only members with status active may use the system
 - A person may self-register, creating an active member dated today and issuing them a member ID

Lending

- Loans: Loan is its own entity (the same member may borrow the same copy on separate occasions)
 - Loan ID
 - Checkout date
 - Due date
 - Returned date (empty while the loan is open)
- Holds: relationship between Member and Item, held at title level (member doesn't care which physical copy)
 - Date placed (relationship attribute)
- Fines: weak entity owned by Loan — at most one fine per loan
 - Amount
 - Date issued
 - Status (unpaid, paid)
- Relationships:
 - Loan — Loan Of — Copy
 - Loan — Borrowed By — Member
 - Loan — Incurs — Fine
 - Member — Places Hold — Item
- Loan rules:
 - Loan period is 14 days from checkout
 - A member may hold multiple loans at once, but at most one copy of any given item
 - Borrowing sets the copy's status to on loan automatically
 - Returning sets the returned date to today and restores the copy to available automatically
 - A member may only return a copy against one of their own open loans
 - Loan history is retained after return — rows are never deleted
- Hold rules:
 - A member may hold a given item at most once
 - A hold may only be placed when no copy of that item is available — otherwise the member is told to borrow it

- Fine rules: applied to overdue loans
 - Fine amount depends on days late
 - Fines on unpaid, still-overdue loans are recalculated as they grow
 - Unpaid total above \$20 blocks new borrows

Donations

- Members may donate items to the library
- Relationship: Member — Donates — Copy (a donation always produces a specific copy)
 - Date donated (relationship attribute)
- Rules:
 - Supported for Print Books, EBooks, Magazines and Journals
 - If the item already exists (matched by ISBN for books, ISSN + issue for magazines and journals), the donation adds a new copy instead of duplicating the catalogue entry
 - If the item is new, a new Item, its type record, and copy number 1 are created together
 - Donated copies enter the collection as available

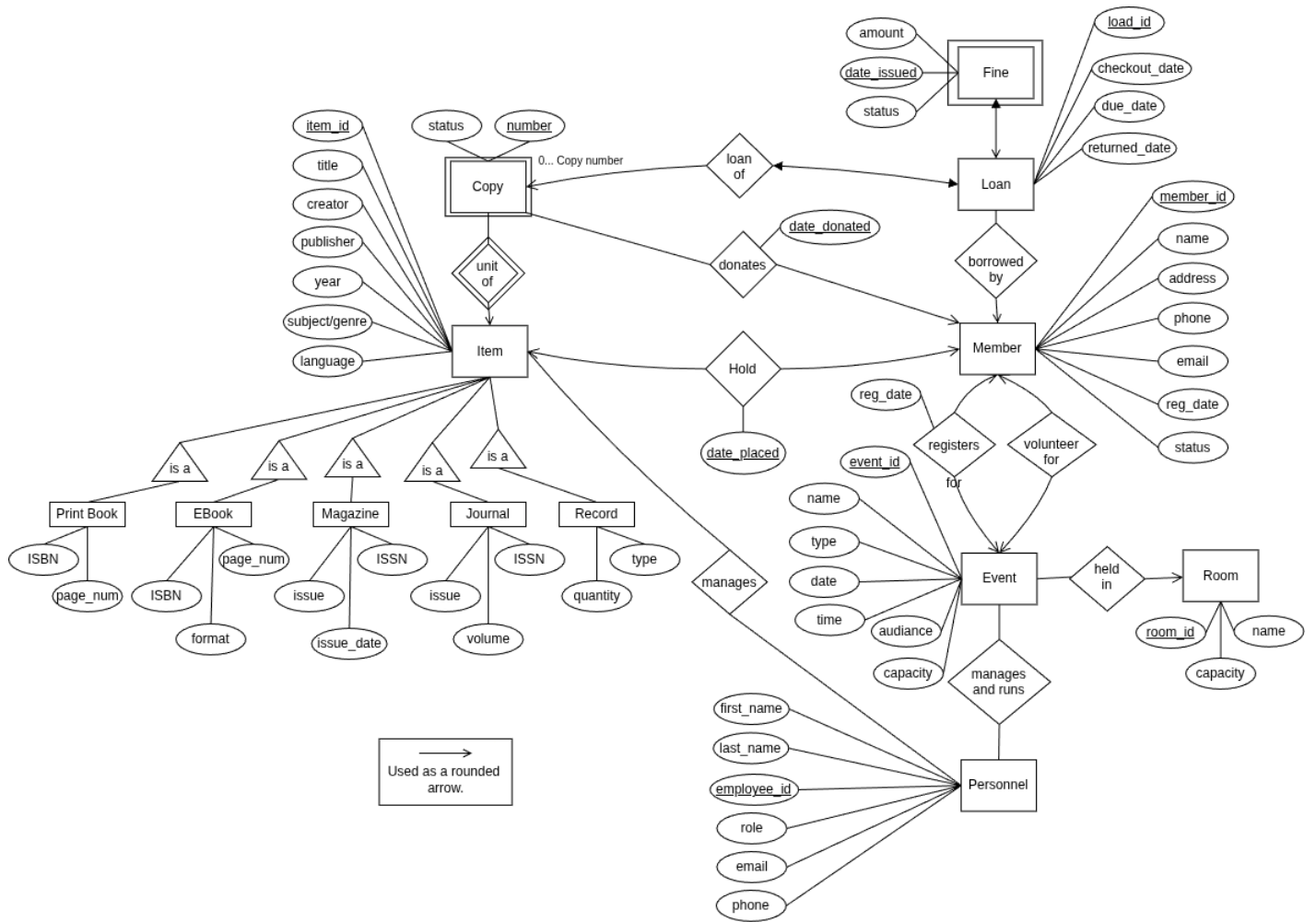
Events

- Types: Book Club, Art Show, Film Screening, Author Talk, Workshop
- Attributes:
 - Event ID
 - Name
 - Type
 - Date
 - Time
 - Recommended audience (kids, teens, adults, all ages)
 - Capacity (max attendees)
- Rooms: Room is its own entity
 - Room ID
 - Room name
 - Capacity

- Relationships:
 - Event — Held In — Room
 - Event — Managed And Run By — Personnel (Event Coordinator for general events, Curator for exhibits)
 - Member — Registers For — Event (with registration date as a relationship attribute)
 - Member — Volunteers For — Event
- Rules:
 - Attendance is free (no payment/attribute needed)
 - A member may register for, and volunteer at, a given event at most once each
 - Registration is refused once registrations reach the event's capacity
 - Members browse upcoming events only — those dated today or later

Personnel

- Roles:
 - Chief librarian (oversees whole library/system)
 - Librarian
 - Library staff
 - Frontline staff
 - Digital/electronic resources librarian (manages digital resources)
 - Curator (manages collections/exhibits)
 - Event coordinator
 - Serials librarian (managing journals, magazines etc.)
 - Library IT manager
 - Library IT staff
- Info tracked per employee:
 - Employee ID
 - First name
 - Last name
 - Role
 - Email
 - Phone number
- Relationships:
 - Personnel — Manages — Item (restricted by role: serials librarian manages magazines and journals, digital resources librarian manages EBooks)
 - Personnel — Manages And Runs — Event
- Rules:
 - Members can look up librarian contact details to ask for help



Anomalies

Item(item_id, title, creator, publisher, year, subject_genre, language) FD: item_id $\rightarrow$ title, creator, publisher, year, subject_genre, language item_id* = all attributes $\rightarrow$ superkey **In BCNF**

PrintBook(item_id, ISBN, page_num) item_id $\rightarrow$ item FD: item_id $\rightarrow$ ISBN, page_num item_id* = {item_id, ISBN, page_num} = all attributes $\rightarrow$ superkey FD: ISBN $\rightarrow$ item_id, page_num ISBN* = {ISBN, item_id, page_num} = all attributes $\rightarrow$ superkey **In BCNF**

EBook(item_id, ISBN, format) item_id $\rightarrow$ item FD: item_id $\rightarrow$ ISBN, format item_id* = {item_id, ISBN, format} = all attributes $\rightarrow$ superkey FD: ISBN $\rightarrow$ item_id, format ISBN* = {ISBN, item_id, format} = all attributes $\rightarrow$ superkey **In BCNF**

Magazine(item_id, ISSN, issue, issue_date) item_id $\rightarrow$ item FD: item_id $\rightarrow$ ISSN, issue, issue_date item_id* = all attributes $\rightarrow$ superkey FD: (ISSN, issue) $\rightarrow$ item_id, issue_date (ISSN, issue)* = all attributes $\rightarrow$ superkey ISSN alone is not a determinant **In BCNF**

Journal(item_id, ISSN, issue, volume) item_id → item FD: item_id → ISSN, issue, volume
item_id* = all attributes → superkey FD: (ISSN, volume, issue) → item_id (ISSN, volume,
issue)* = all attributes → superkey ISSN alone is not a determinant **In BCNF**

Record(item_id, type, quantity) item_id → item FD: item_id → type, quantity item_id* = all
attributes → superkey **In BCNF**

Copy(item_id, copy_number, status) item_id → item FD: (item_id, copy_number) → status
(item_id, copy_number)* = all attributes → superkey **In BCNF**

Loan(loan_id, checkout_date, due_date, returned_date, item_id, copy_number, member_id)
(item_id, copy_number) → Copy, member_id → Member FD: loan_id → checkout_date,
due_date, returned_date, item_id, copy_number, member_id loan_id* = all attributes →
superkey FD: (item_id, copy_number, checkout_date) → loan_id (item_id, copy_number,
checkout_date)* = all attributes → superkey **In BCNF**

Fine(loan_id, date_issued, amount, status) loan_id → Loan FD: loan_id → date_issued,
amount, status loan_id* = all attributes → superkey **In BCNF**

Member(member_id, name, address, phone, email, reg_date, status) FD: member_id →
name, address, phone, email, reg_date, status member_id* = all attributes → superkey FD:
email → member_id, name, address, phone, reg_date, status email* = all attributes →
superkey **In BCNF**

Hold(member_id, item_id, date_placed) member_id → Member, item_id → Item FD:
(member_id, item_id) → date_placed (member_id, item_id)* = all attributes → superkey **In BCNF**

Donates(item_id, copy_number, member_id, date_donated) (item_id, copy_number) →
Copy, member_id → Member FD: (item_id, copy_number) → member_id, date_donated
(item_id, copy_number)* = all attributes → superkey **In BCNF**

Room(room_id, name, capacity) FD: room_id → name, capacity room_id* = all attributes →
superkey **In BCNF**

Event(event_id, name, date, room_id, type, time, audience, capacity) room_id → Room FD:
event_id → name, date, room_id, type, time, audience, capacity event_id* = all attributes →
superkey FD: (room_id, date, time) → event_id (room_id, date, time)* = all attributes →
superkey **In BCNF**

Registers(member_id, event_id, reg_date) member_id → Member, event_id → Event FD:
(member_id, event_id) → reg_date (member_id, event_id)* = all attributes → superkey **In BCNF**

Volunteers(member_id, event_id) member_id → Member, event_id → Event All-key, no
non-trivial FDs **In BCNF**

Personnel(employee_id, first_name, last_name, role, email, phone) FD: employee_id →
first_name, last_name, role, email, phone employee_id* = all attributes → superkey **In BCNF**

Manages(employee_id, item_id) employee_id → Personnel, item_id → Item All-key, no non-trivial FDs **In BCNF**